Data and Information Management 9 (2025) 100050

Contents lists available at ScienceDirect

Data and Information Management

journal homepage: www.journals.elsevier.com/data-and-information-management

Data use policies on state COVID-19 dashboards in the United States: Key characteristics, topical focus, and identifiable gaps

Rong Tang *, Zhan Hu, Yishan Zhang

School of Library and Information Science, Simmons University, 300 The Fenway, Boston, MA, 02115, USA

ARTICLE INFO

Keywords:
Data use policy
US State COVID-19 dashboards
Open data policy analysis model
Open data principles
FAIR principles

ABSTRACT

In this paper, we report the findings of an investigation into the data use policies published on the COVID-19 dashboards developed by the 50 state governments of the United States as well as the government of District of Columbia. Specifically, we examined the key attributes of the dashboard data notes, such as data source, update frequency, and data suppression disclaimers. We also studied the terms and phrases used, as well as topic themes of the data policy texts. Using a data policy analysis model, our results revealed a series of gaps and inconsistencies in the policy statements. Connecting these gaps and inconsistencies with potential problems that could violate individual Open Data Principles (ODP) and the FAIR principles, we made recommendations to help resolve these missing areas and fix the inconsistencies, so that open government data can be managed and used to further the very core of open data practice. Further research that we plan to carry out includes confirmation and validation of our analysis model and our approach of linking the examination and assessment of open data policy with ODP and the FAIR principles.

1. Introduction

During the COVID-19 pandemic, within the United States, various states produced dashboards tracking COVID-19 development and vaccination progress. As indicated by Shah (2020), "Covid-19 has shown the importance of dashboards and visualization for everyone. Various countries and even states/regions have created their own COVID-19 dashboards to educate people on the pandemic." Curioso and Carrasco-Escobar (2020) pointed out the crucial role of open data during a pandemic situation: "Local, regional and national governments play a key role in this pandemic …. move towards open data will be essential to face this pandemic, contributing to better transparency, reproducibility of results and evidence-based decision-making" (p. 1). Matheus et al. (2020) also provided a long list of benefits of using dashboards, including providing an overview at a glance, having transparency and accountability, visualizing trends, and "disclos[ing] relevant information to people" (p. 5). Furthermore, the authors mentioned that there are a greater number of risks and challenges, including low data quality, lack of resources, "no maintenance and updating," data privacy issues, and "scarce usage" (p. 6). Given the challenges faced by local governments in developing and maintaining open data dashboards, it would be crucial for dashboard users, including researchers and everyday citizens, to understand the limitations of dashboards datasets. Such limitations should be articulated in a data use policy. This is particularly true for the COVID-19 data dashboards developed by various state governments in the United States for the purpose of providing timely information. These dashboards need to provide a data policy statement to outline key facts associated with the data presented, including data source, data update frequency, term definitions, data scope, data suppression disclaimer, and more. This is what motivated the present study with its focus on published data notes related to the use of dashboards' data.

Over the past years, a number of research studies investigated usage habits (e.g., Röddiger et al., 2020) and user experience with map-based interaction with COVID-19 dashboard visualizations (e.g., Cay et al., 2020). However, seldom has any study investigated the data policy statements/disclaimers published on open access dashboards. The present study examines the data use policies published on the COVID-19 dashboards developed by state governments across the United States. Guided by Open Data Principles (Project Open Data, 2010) and the FAIR principles (Wilkinson et al., 2016), as well as the framework for comparing open data policies at different government levels developed by Zuiderwijk and Janssen (2014), the present study examined the major characteristics such as data source, data suppression disclaimer, as well as the content of the data use policy.

* Corresponding author.
E-mail address: rong.tang@simmons.edu (R. Tang).

https://doi.org/10.1016/j.dim.2023.100050
Received 25 April 2023; Received in revised form 1 August 2023; Accepted 8 September 2023
2543-9251/© 2023 Published by Elsevier Ltd on behalf of School of Information Management Wuhan University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

The three key research questions that we pursue are as follows. While RQs 1 and 2 address the content characteristics of the data policy notes, RQ3 delves into the observable gaps in these policy notes which lead to the mismatch with the ODP and FAIR guidelines.

RQ1. What are key attributes characterizing a published data use policy or data note on COVID-19 dashboards developed by various states in the United States?
RQ2. What are the most frequently occurring terms, phrases, and key topical themes in the data policy texts?
RQ3. What notable gaps and inconsistencies may be identified in the open data policies manifested through state developed/sponsored COVID-19 dashboards? What mismatches do these gaps induce leading to problems with achieving the Open Data Principles and the FAIR Principles?

2. Literature review

2.1. Citizens' use of open data

With the advancement of open government and open data, citizens have been found to frequently access data through government portals (Horrigan & Rainie, 2015). In their study back in 2010, Smith (2010) found that, in general, a higher proportion of citizens in the United States dealt with local government (52%) in their everyday situation than with state government (22%) or federal government (15%). When it comes to using the datasets from government sites, 16% visited a site such as data.gov, recovery.gov or usaspending.gov. Note that Smith's report indicated that "40% of online Americans have gone online for data" (p. 3) about the government; five years later, in a survey of 3212 American residents, Horrigan and Rainie (2015) found an increase in the proportion: "65% of Americans in the prior 12 months have used the internet to find data or information pertaining to the government" (p. 3). In addition, the authors found that "Americans are for the most part comfortable with the government sharing online data about their communities" (p. 6), although a much smaller proportion of people were comfortable with the government sharing their specific home mortgage information.

Meanwhile, a few studies looked into the specific use scenarios for citizens accessing government datasets. According to Davies (2010), there are five types of use cases: (1) data to fact, (2) data to information, (3) data to interface, (4) data to data, and (5) data to service. The specific data activities for each use scenario vary. For example, "data to fact" involves the individual users who "seek out specific facts in a newly open dataset" (p. 2), whereas "data to information" describes a situation in which the users transform data into visualizations, reports, and graphics.

2.2. Data use policy, open data principles, and the FAIR principles

Over the years, there have been plentiful studies on open government datasets (e.g., Nayek, 2018; Tang & Jiang, 2021; Wang et al., 2019; Zuiderwijk & Reuver, 2021), with the investigative focus mostly on the characteristics of the datasets on OGDs, such as data format, data categories, and data levels. Seldom has any study examined the data use policies of open datasets. With citizens being able to access government datasets, it is essential that they are made aware of the delimiters associated with those datasets. Two specific papers, one by Goodspeed (2011) and one by Zuiderwijk and Janssen (2014), paid attention to the particular content of the data policy. In analyzing 42 data disclaimers, licenses, and policies of municipalities in Massachusetts for the use of their open GIS data, Goodspeed (2011) identified eight common provision items included in these statements. The most common provision was "Data provided without warrantee," followed by "Municipality indemnified and held harmless," "Restrictions on reuse," "No claim to data updates," and "Data not for legal purpose." The author further discussed the challenges and obstacles in maintaining open government data managed by local governments with limited staffing and resources. Goodspeed (2011) pointed out that "Open government will require not only moving beyond a narrowly conceived 'right to view' records but also an ongoing discussion of a range of practical, ethical, and legal considerations such as file formats and distribution technology, balancing openness with privacy and public safety considerations, and appropriate data policies and licenses "(p.30).

In their study, Zuiderwijk and Janssen (2014) examined the open data policies of seven Dutch governmental organizations using a framework of analysis based on their literature review. Their framework includes a section on "Policy environment and context," then a section on "Policy content (input)," which leads to "performance indicators (output)" and finally "Public values (impact)." For "Policy content," there were 17 elements, ranging from "Policy strategy and principles for opening data," to "Policy measures and instruments," "Processing of data before publication," "Amounts of opened data," "Types of open data," "Way of presenting data," and more. The authors' analysis revealed differences "with regard to policy objectives, policy instruments, principles for opening data, the amounts and types of publicized and unpublicized data, the metadata that is provided with the open data, support for the use of the open data, the types of usage of publicized data, and expectations surrounding the positive effects and the risks" (p. 27). The authors urged governmental organizations to collaborate to develop open data policies, and they further declared that "As the impact of open data policies becomes visible over time, we recommend developing measures for evaluating the impacts of open data and using these measures to evaluate the effects of open data policies." (p. 28).

The development of open data policies should be guided by established open data standards or principles. In the context of the United States, back in 2009, the Obama administration White House, 2009 issued a "Memorandum on Transparency and Open Government," which declared that the government should be transparent, participatory, and collaborative. A set of open data principles was established by the US "Open Government Directive" (Horrigan & Rainie, 2015). According to *Project Open Data,* which is under the Directive, open government data should be *Public, Accessible, Described, Reusable, Complete, Timely,* and *Managed post-release.* The detailed description of each of these seven principles is provided in Table 1. Parallel to these principles, in the realm of scientific research, the FAIR principles were published as the guidelines for "improv[ing] the Findability, Accessibility, Interoperability, and Reuse of digital assets." (https://www.go-fair.org/fair-principles/). There are overlaps between the Open Data Principles and the FAIR principles, and Table 1 attempts to create a crosswalk between the two sets of principles. It is clear that several key principles are repeated between the two; however, the unique principles are: *Public, Complete, Timely, Managed post-release* from Open Data Principles, and *Interoperable* from FAIR. Combining the open data principles with the FAIR principles, we have a comprehensive and applicable set of guidelines for the elements that should be included in the content of an open data use policy.

Meanwhile, specifically pertaining to Open Government Data, similar sets of principles were developed by various groups of advocates and government policy experts. In 2007, 30 OGD experts gathered in California and formulated eight principles of open government data, which include: *Complete, Primary, Timely, Accessible, Machine processable, Non-discriminatory, Non-proprietary,* and *License-free* (Tauberer, J. n.d.). This set of principles were enhanced by the Sunlight Foundation. Currently, countries from Europe, such as Austria, list ten OGD principles (Eibl, et al., 2019) as the following: *Completeness; Primary source; Timely provisioning; Easy access; Machine readable; Freedom from discrimination; Use of open standards; Licensing; Documentation (durability);* and *Utilization costs.* One can see a lot of overlap between the OGD principles and Seven Open Data Principles listed in Table 1. Connected with the ODP and OGD principles, W3C's (Lóscio et al., 2017) "Data on the Web Best Practices" also provides examples of best practices associated with data. The document lists the challenges of publishing or consuming data on the Web, and it also outlines the benefits of using the best practices as the following: *Comprehension; Processability; Discoverability; Reuse; Trust;*

Table 1
Crosswalk between open data principles and FAIR guidelines (sources: Open data Project; Horrigan & Rainie, 2015; go-fair.org).

	Open Data Principles	FAIR
Public	*Agencies should have a presumption that their data should be open, subject to laws and imperatives of privacy, confidentiality, security, or other valid restrictions.*	
Accessible	*Data should be machine readable and easily retrievable and searchable.*	*A1. (meta)data are retrievable by their identifier using a standardized communications protocol* *A2. metadata are accessible, even when the data are no longer available*
Described → Findable & Reusable	**Described** *Open data should be clearly described so that users can understand how to use them, as well as the data's strengths and weaknesses.*	**Findable** *F1. (Meta)data are assigned a globally unique and persistent identifier* *F2. Data are described with rich metadata* *F3. Metadata clearly and explicitly include the identifier of the data they describe* *F4. (Meta)data are registered or indexed in a searchable resource* **Reusable** *R1. (Meta)data are richly described with a plurality of accurate and relevant attributes*
Reusable	*Under open licenses, there should be no restrictions on use.*	*R1. (Meta)data are richly described with a plurality of accurate and relevant attributes* *R1.1. (Meta)data are released with a clear and accessible data usage license* *R1.2. (Meta)data are associated with detailed provenance* *R1.3. (Meta)data meet domain-relevant community standards*
Complete	*Data should be published in primary forms.*	
Timely	*Data should be released as quickly as necessary to preserve its value.*	
Managed post-release	*Agencies should designate a point-of-contact for data releases to respond to queries.*	
Interoperable		*I1. (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation.* *I2. (Meta)data use vocabularies that follow FAIR principles* *I3. (Meta)data include qualified references to other (meta)data*

Linkability; Access; Interoperability. Note also the overlapping elements such as discoverability, reuse, and interoperability with both ODP and the FAIR principles. From the data consumer's perspective, the benefits of discoverability and reuse are described as "machines will be able to automatically discover a dataset or data within a dataset," and "the chances of dataset reuse by different groups of data consumers will increase."

With much overlap between a variety of established principles, a combination of ODP and FAIR guidelines were used to assess data policy statement content on COVID-19 dashboards in the U.S.

3. Data policy analysis model

To operationally document and examine the content of open data policy statements, there is a need for a suitable and feasible analytical framework. Based on the literature, specifically guided through the open GIS data disclaimer provision items found by Goodspeed (2011) and elements of open data policy content outlined in Zuiderwijk and Janssen (2014), and through our preliminary examination of the date notes on COVID-19 U.S. state dashboards, the research team developed a data policy analysis model (see Fig. 1) which measures various elements considered essential for data use statements. For instance, the element of "data update frequency" in this analysis model is based on the "no claims to data updates" finding from Goodspeed (2011), and the element of "data subject to change" is connected with the "Data provided without warrantee" of the same article (p. 27). The paper by Zuiderwijk and Janssen (2014) also provided several inspirations, including, but not limited to, the "data scope" element being linked with the "amount of open data" item, and the "data suppression" element being associated with the "Types of data not publicized" item in the policy content category of the open data policy comparison framework (p.20). Incorporating these elements into a data policy statement will fulfill various open data principles such as findable, reusable, complete, timely, described, accessible, and more. Here we first present the data policy analysis model that specifies on individual policy elements such as data sources. In the Results section, we report the findings regarding each element of the data policy statements. Subsequently, through the analysis of the gaps and inconsistencies of existing data policy statements, we identify the negative impact of these problems, causing the violations in following the ODP and FAIR standards. We then make recommendations for addressing the problems.

4. Methods

4.1. Data collection

The focus of our study is on the data use policies published on the COVID-19 dashboards developed by state government across the United States. Specifically, we examined the key attributes of data policy statements, such as data source, data suppression disclaimer, as well as the term and phrase frequency, and the topic clusters of the policy text. The research team started collecting and coding data from October 2022 to February 2023. We first built an inventory of the dashboards of 50 states and the District of Columbia. In data harvesting and coding, we paid

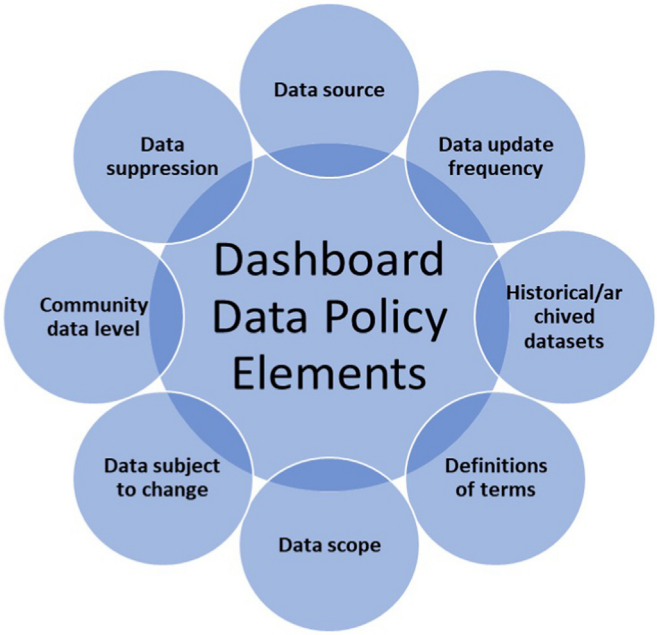

Fig. 1. Data policy analysis model.

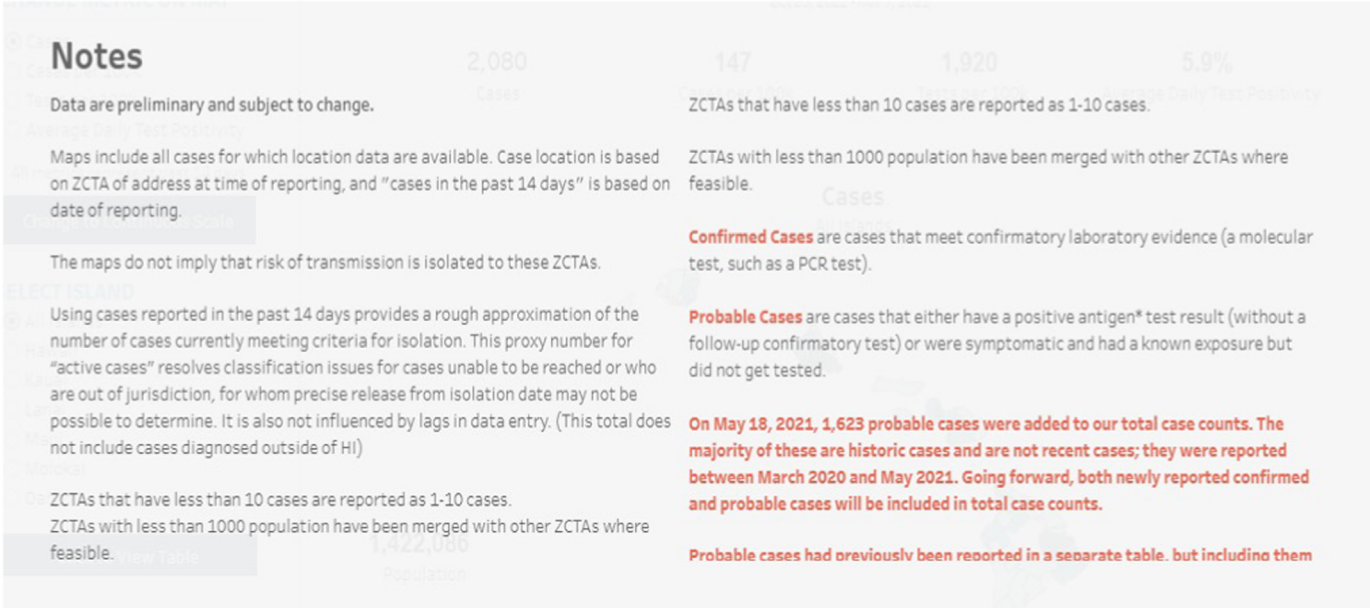

Fig. 2. Hawaii COVID-19 data notes.

specific attention to extracting the content of the data policy statement. For instance, from the Hawaii COVID-19 Data Dashboard (https://health.hawaii.gov/coronavirusdisease2019/current-situation-in-hawaii/), we captured the content of data notes, as shown in Fig. 2.

The placement of the data notes varied from being a separate page or tab to being embedded in the dashboard as shown in Fig. 3, the Coronavirus - Michigan Data Dashboard (https://www.michigan.gov/coronavirus/stats) screen.

4.2. Data coding

For the 51 dashboards, we examined whether there were data notes or disclaimer statements about data policy. The research team identified where the data notes appeared and recorded the original texts. In cases where the data notes were part of the images generated by the platform, such as Tableau, we used tools to convert the images into text. The recording of the presence or absence of certain item (e.g., data suppression disclaimer) was through a Google sheet where the research team tallied the frequency of the occurrences and the characteristics of certain items when the recording was done.

In this study, we treat both the separate and explicit "data notes" section, a link to a separate page/file, and the explanatory notes embedded on the dashboard (underneath a chart, or under the label) as part of what we consider the data notes.

4.3. Data analysis

For data analysis, we coded the "data policy" text excerpts with aspects of the proposed data disclaimer element framework. Then, descriptive analysis was performed in the Google sheet, term and phrase frequency analysis with the Voyant tools (Sinclair & Rockwell, 2016), and topic modeling with the jsLDA tool (Mimno, 2013). The Voyant tools is a widely used "web-based text reading and analysis environment" (Voyant Tools, n.d.) which generates "text analytics such as word frequency lists, frequency distribution plots, and KWIC displays" (Klein et al., 2015, p.i138). Meanwhile, the jsLDA tooljsLDA, is a browser-based topic modeling tool developed by Professor David Mimno from Cornell University through running Latent Dirichlet allocation (LDA) topic models using JavaScript. According to a reviewer, compared to the Voyant tools where topic modeling is mostly hidden, (jsLDA,) is "clearly designed to facilitate topic modeling," and has "the ability to explore connections between topics" (Dombrowski, 2022).

5. Results

5.1. Key attributes of COVID-19 dashboard data policy

5.1.1. Data policy length

Among the 51 dashboards, all but two (Kentucky and Alaska) had published data policy statements. For the valid texts we used for our analysis, the average length of the data policy statements was 225.76 words, with a minimum length of 13 and maximum length of 1377 words. Iowa COVID-19 Reporting Dashboard[1] (https://idph.iowa.gov/Emerging-Health-Issues/Novel-Coronavirus/COVID-19-Reporting) had the shortest policy statement, "All data presented are provisional and may change as additional reports are received," whereas Idaho COVID-19 Dashboard (https://public.tableau.com/app/profile/idaho.division.of.public.health/viz/DPHIdahoCOVID-19Dashboard/Home) had the longest policy statement.

5.1.2. Data source

Among the 51 dashboards, 49 explicitly mentioned their data source in the data policy notes, while 2 did not include the data source in their policy statement, but we were able to trace it from outside the policy statement. For all 51 dashboards, the state department of health or public health or its variations was credited as responsible for collecting or providing data to the dashboard.

5.1.3. Data update frequency

Among the 51 dashboards, all but one had update frequency information. Of those 50 dashboards, the most common update frequency was weekly ($n = 39$, 78.00%), followed by daily ($n = 9$, 18.00%) and bi-weekly ($n = 2$, 4.00%). Not all the dashboards' update frequency information is placed in an obvious or consistent position.

[1] Note that as of April 25, 2023, the Iowa's Dashboard was no longer active. What was analyzed and reported in this paper was the dashboard that remained active till February 2023 and whose content was captured by the research team.

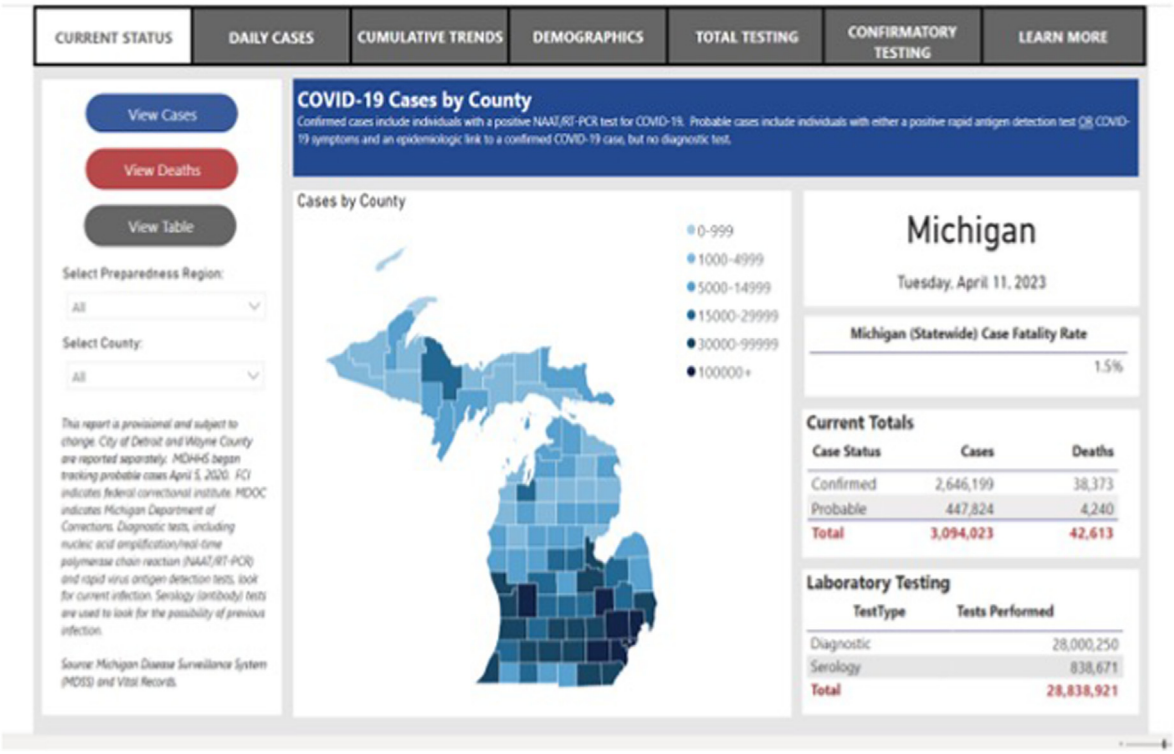

Fig. 3. Michigan COVID-19 Dashboard Screen (embedded data notes).

5.1.4. Historical/archived datasets

Among the 51 dashboards, 17 (33.33%) included information about the handling of historical/archived datasets. The majority (n = 34, 66.67%) of the dashboards had no mention of past datasets outside the date range the dashboard displays, either on the dashboard itself or in the data notes section. Dashboards by South Carolina (https://scdhec.gov/covid19/covid-19-data) and Massachusetts (https://www.mass.gov/info-details/covid-19-response-reporting#covid-19-interactive-data-dashboard-) contained information about archived data and had a link to a separate section under the dashboard providing links to reports and raw data of historical COVID-19 cases. Meanwhile, dashboards by Connecticut (https://data.ct.gov/stories/s/q5as-kyim) and Kansas (https://www.coronavirus.kdheks.gov/160/COVID-19-in-Kansas) had a link to a separate page that served as an archive of PDF reports on COVID-19 from the state Department of Public Health.

5.1.5. Definitions of terms

Among the 51 dashboards, 38 (74.51%) had term definitions either in their data notes section or embedded on the dashboard, whereas 13 (25.49%) dashboards had no clear definitions of terms. The defined terms included "test positivity," "hospitalized patients," "reinfection," "imported cases," and "probable deaths."

5.1.6. Data scope

We also examined whether there were notes describing the scope of the data reflected. Among the 51 dashboards, 23 (45.10%) had some explanation or disclaimer about data coverage, whereas a majority (n = 28, 54.90%) of the dashboards did not have any scope description. Those that provided the data scope description indicated whether the data covered statewide data or a statement such as "COVID-19 negative test result data is not yet fully integrated into the Public Health systems" from the New Hampshire COVID-19 Response Dashboard (https://www.covid19.nh.gov/dashboard).

5.1.7. Data subject to change disclaimer

Among the 51 dashboards, a majority (n = 43, 84.31%) had a disclaimer on the fact that data is provisional and subject to change, whereas eight (15.69%) dashboards did not have such a statement. Examples of the subject to change disclaimer were statements such as "All data displayed is preliminary and subject to change as more information is reported to ODH" from Ohio's COVID-19 Dashboard (https://coronavirus.ohio.gov/dashboards/overview), "All data presented are provisional and subject to change as additional reports are received" from Missouri's COVID-19 Data Dashboard (https://health.mo.gov/living/healthcondiseases/communicable/novel-coronavirus/data/public-health/), or "Data are preliminary and subject to quality improvement and quality assurance validation" from the COVID-19 Cases in Kansas Dashboard (https://www.coronavirus.kdheks.gov/160/COVID-19-in-Kansas).

5.1.8. Community data level

Among the 51 dashboards, only 17 (33.33%) indicated that the dashboard county map followed the CDC-defined community risk levels or that they referred to the CDC's definition, whereas a majority (n = 34, 66.67%) of the dashboards did not include such information. For example, the DC's COVID-19 Surveillance Dashboard (https://coronavirus.dc.gov/data) states that "District of Columbia monitors key public health signs of multiple communicable diseases as a part of routine surveillance activities and disseminates health communications to decrease community spread. The following subset of key metrics, aligned with CDC's Community Levels, can be used in conjunction with wider public health messaging to support residents in personal risk decision-making."

5.1.9. Data suppression disclaimer

Among the 51 dashboards, only 11 (21.57%) of them stated that they had some forms of data suppression in order to protect residents' privacy, whereas a majority (n = 40 78.43%) of the dashboards did not include such an indication. For example, Maryland COVID-19 Data Dashboard (https://msdh.ms.gov/msdhsite/_static/14,0,420.html#Mississippi) indicated that "Data for ZIP codes with 7 or fewer cases are suppressed." Connecticut COVID-19 Update Dashboard (https://data.ct.gov/stories/

s/q5as-kyim) stated that "To protect confidentiality, the data in towns with fewer than five (5) cases or five (5) positive NAAT tests in the past seven (7) days are suppressed."

5.2. Terms and phrases used in data notes

Out of 51 dashboards, 49 (96.08%) had some textual content in the data notes either placed separately ($n = 26$, 53.6%) or embedded ($n = 17$, 34.69%) in the dashboard underneath the charts or caption of the charts. There were also six (12.24%) dashboards where the data notes appeared both outside the dashboard and embedded on the dashboard.

Data notes on dashboards also vary in length and level of detail. The majority of the data notes included definitions of terms such as "confirmed cases." They also included update frequency, data source, and disclaimer of data scope. Out of 49 dashboards with data notes, 47 (95.92%) listed their data source. All data sources were from a department within the state government; for instance, the data source listed in the COVID-19 in Mississippi Dashboard (https://msdh.ms.gov/msdhsite/_static/14,0,420.html#Mississippi) was "Source: Mississippi State Department of Health."

5.2.1. Term frequency

When combining 49 data policy statements, the Voyant tool revealed that there were a total of 10,300 words and 1594 unique terms. The top ten most frequent terms were: data ($n = 334$, 3.243%); case ($n = 179$,1.738%); report ($n = 128$, 1.243%); covid ($n = 119$, 1.155%); information ($n = 82$, 0.796%); health ($n = 81$, 0.786%); death ($n = 65$, 0.631%); dashboard ($n = 62$, 0.602%); day ($n = 61$, 0.592%); update ($n = 60$, 0.583%); and test ($n = 60$, 0.583%).

The term "data" was found significantly correlated with *cov* ($r = -0.78$, $p = 0.008$), coronavirus ($r = -0.74$, $p = 0.014$), collected ($r = -0.713$, $p = 0.021$), cases ($r = -0.713$, $p = 0.021$), changes ($r = -0.713$, $p = 0.021$), counts ($r = -0.713$, $p = 0.021$), classified ($r = -0.713$, $p = 0.021$), availability ($r = -0.713$, $p = 0.021$), booster ($r = -0.713$, $p = 0.021$), cli ($r = -0.713$, $p = 0.021$), compatible ($r = -0.71$, $p = 0.021$), creation ($r = -0.71$, $p = 0.021$), age ($r = 0.71$, $p = 0.023$), conducted ($r = -0.69$, $p = 0.026$), contain ($r = 0.68$, $p = 0.029$), 100,000 ($r = -0.66$, $p = 0.0387$), and created ($r = -0.64$, $p = 0.047$).

5.2.2. Phrase analysis

The examination of phrases in the data policy notes on dashboards was carried out through two aspects using the Voyant tool: (1) all phrases included, and (2) phrases that start with the term "data." Knowing the most frequently occurring all-phases or data-anchored phrases help us to understand the common practices in data policy notes across COVID-19 dashboards of varying states. It also potentially leads to the identification of best practices or moving towards standardization of data policy for COVID-19 dashboards or for dashboards containing open public health data.

The top three most commonly occurring phrases among all phrases were: covid 19 dashboard/covid 19 data dashboard ($n = 8$, 1.34%), emergency department visit/s ($n = 6$, 1.01%), and on the dashboard ($n = 5$, 0.84%). Meanwhile, the top three data-anchored phrases were: data source ($n = 13$, 5.80%), data are/is provisional ($n = 12$, 5.36%), and data are preliminary ($n = 11$, 4.91%). Figs. 4 and 5 display the most frequently occurring among all phrases and among data phrases.

5.3. Themes from topic modeling

When importing the entire policy statement texts into the jsLDA tool, 8 topic clusters emerged. Fig. 6 lists those topic themes with some keywords highlighted in red. The topic clusters ranged from "report and chart based on office data resident sample contained in the map," "population estimate use calculated average demographic groups such as age group, low number data were suppressed for individual privacy," to "Dashboard data are provisional or preliminary and subject to change, case report updated weekly," and more. As shown in Fig. 6, some of the topic clusters match with the areas that we reported earlier in terms of data suppression, data update frequency, data sources, community levels, and data are subject to change.

5.4. Inconsistencies and gaps

Overall, there are a great number of inconsistencies in how data policies were presented across the 51 dashboards. First and foremost, the placement of data policy statements was inconsistent. There were also significant variations in terms of the level of depth and detail of the policy

Fig. 4. Most frequently occurring phrases in dashboard data policy statements.

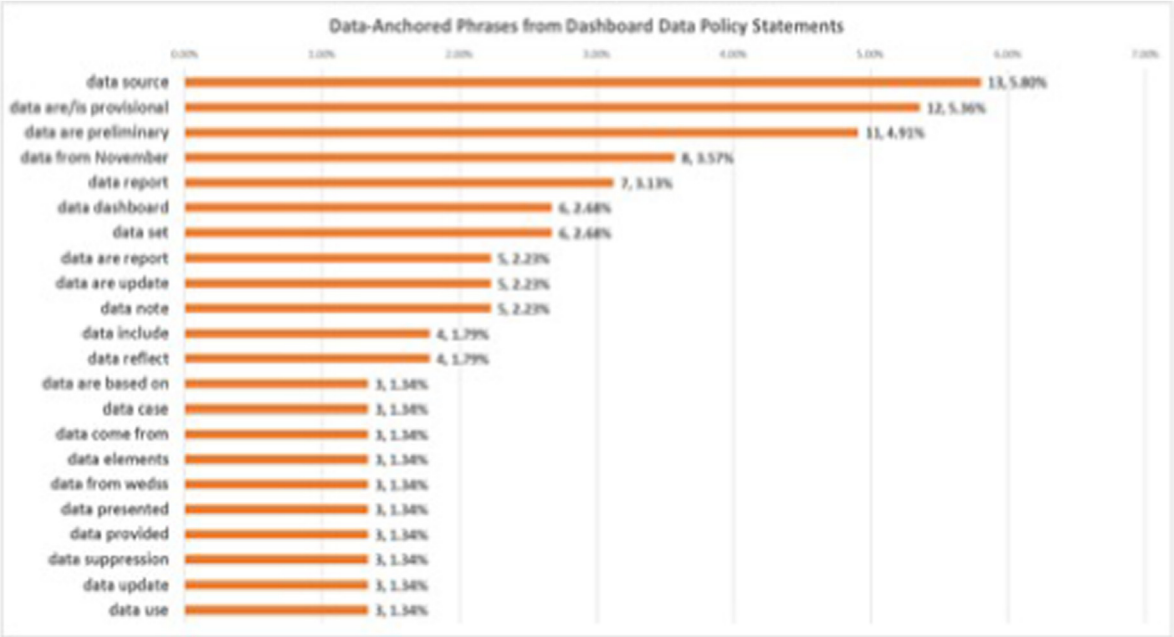

Fig. 5. Data-anchored phrases from dashboard data policy statements.

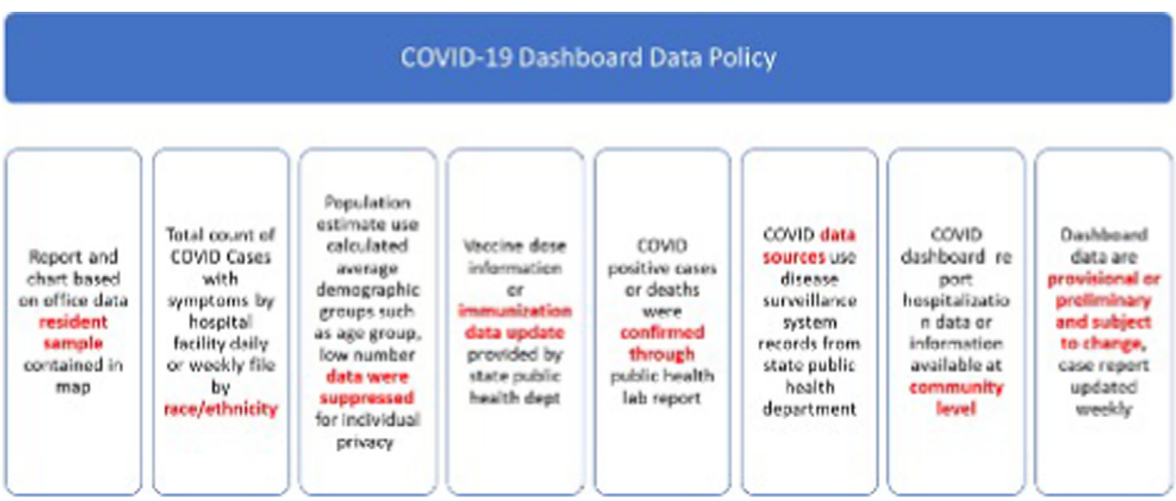

Fig. 6. Topic clusters derived from topic modeling of policy statement texts.

statement and elements addressed. Another problem is that even if a given element such as data update frequency is included, the elements were scattered or blended in different parts of the dashboards or subsequent websites. Where a dashboard is a suite of sub-dashboards, the policy content might appear in different sub-dashboards, and each sub-dashboard might have its own policies. In the following sections, we present inconsistencies or gaps from different aspects of the data policy statements. When appropriate, we also make recommendations for best practices concerning specific aspects of the data policies.

5.4.1. Data source

While most of the dashboards credited the state department of health/public health as the organization that collected, aggregated, and presented the data for the dashboard, variation occurred between dashboards that included information on the origin of certain datasets (e.g., hospital, secure information system) and others that did not release the details of sources of the original datasets. To give full disclosure and provide accurate reports, the data policy should list not only the organization that was responsible for aggregating and displaying the data, but also all organizations where datasets were gathered from.

5.4.2. Data update frequency

While all but one of the dashboards included update frequency information, it was inconsistent as to where the update frequency information was located on the dashboard. Additionally, when it comes to a dashboard set consisting of multiple sub-dashboards (e.g., COVID-19 cases or death, hospitalization, vaccination), it was not always clear whether the update frequency varied based on which sub-dashboard the user looked at. Furthermore, the update frequency changed through the development of the pandemic, moving from very frequent updates during the heyday of the virus to less frequent updates, and even to cases in which the updates discontinued. Given the dynamics of the pandemic, all update frequency should be clearly documented and placed in an area that is accessible.

5.4.3. Historical/archived datasets

Less than 35% of the dashboards contained information about historical datasets. With the absence of information on archived datasets, users may not have an accurate and complete understanding of what the datasets entail. It is our belief that all dashboards should not only include information about datasets of past dates, but also take good care in organizing and documenting achieved datasets. Furthermore, there were inconsistencies in the format of historical and archived datasets.

When applying the essence of the open data principles and the FAIR principles to the archived COVID-19 dashboard datasets, it is our understanding that not only should these historical datasets be findable and accessible, but also interoperable in terms of dataset format, and therefore reusable by the general public.

5.4.4. Definitions of terms

Even though close to three-quarters of the dashboards had some terminology definitions, where the definitions appeared varied. The dashboards that had no clear definitions of terms may have caused confusion for their users. The complexity of term definitions also could have added usability and accessibility issues. For those that provided term definitions, not all provided information of the original source for the definition, even though several dashboards referred to the CDC for definitions of selected terms. Additionally, the definitions may evolve over time. It would be helpful to disclose the specific changes that had occurred in the definitions.

It is recommended that the definitions of terms are necessary, and they should be placed in a consistent and obvious location. If definitions are provided as text embedded in different sections of the dashboard, we recommend that a separate and central location of definitions should also be provided in addition to dispersed definition texts.

5.4.5. Data scope

With less than half of the dashboards providing descriptions of the data coverage, users are often left without the understanding of the full scope of the data displayed. It would be helpful for all dashboards to specify what is and is not included in the dashboard, not only in terms of the breadth, but also the depth of the data. Furthermore, in visualizing the trend data, the dashboard should in all instances include time-stamps. The dashboards should also enable users to customize the datasets to the specific scope that they are interested in.

5.4.6. Data subject to change disclaimer

One of the main strengths among dashboards data notes was that more than 80% had a disclaimer of "data is provisional/preliminary and subject to change." This is shown also through our phrase analysis and topic modeling. Nevertheless, more than 15% of the dashboards did not include such a statement. As this is true across all dashboards, we recommend that all dashboards, without exception, explicitly include such a disclaimer.

5.4.7. Community data level

For COVID-19 dashboards, only 33.33% of the dashboards indicated that their county map followed the CDC defined community risk levels, and there were cases where the dashboard presented color coding without indication of whether this was uniquely assigned by the state dashboard or aligned with CDC risk levels. For those cases, the user who is familiar with the CDC levels may make an assumption that this is the same as CDC. It is important to have a statement explaining the risk levels, whether it is per CDC or otherwise.

5.4.8. Data suppression disclaimer

Unfortunately, less than a quarter of the dashboards' data policies included a disclaimer on data suppression. It is helpful for all dashboards to include this aspect of data display. Among those who had a disclaimer, the threshold varied from 5 or 7 to 10. It is also inconsistent as to whether the suppression was by zip codes or by town. And for demographic breakdowns such as gender or ethnicity, the boundary may be different. Furthermore, when the data has been suppressed, it is unclear the outcome of the suppression. Is data still being counted, or reported as 0 cases, or combined with other areas where the cases are more than the threshold number?

6. Discussion and conclusion

In this study, through applying an analysis model, we examined various components of data use policy on 51 U.S. state and District of Columbia COVID-19 dashboards. We discovered the key common characteristics of policy content, and subsequently identified areas of inconsistencies and gaps. In an attempt to connect the missing areas of data policy statements with their potential negative impact on following the open data principles and the FAIR principles, we created a table to show the specific gaps that lead to violations of specific principles (see Table 2).

As shown in Table 2, missing certain elements, inadequacy in description (metadata), and unclear/non-explicit disclaimers would cause a multitude of violations of specific open data principle (ODP) and FAIR principle items. For example, missing original data source information would cause problems in ODP's *Described, Reusable,* and *Complete* and problems with FAIR's *Findable* and *Reusable*. Inconsistent placement of data update frequency information, as well as missing specific sections' update frequency information would make it difficult in meeting the ODP's *Timely* and *Complete*, or FAIR's *Findable* and *Interoperable.* Meanwhile, while the data suppression statements are important to include in the data notes, when the suppression threshold is unclear, together with the outcome of suppression to data counts not specified, it impacts negatively on ODP's *Described*, *Reusable*, and *Complete*, FAIR'S *Interoperable* and *Reusable*. As outlined in the W3C best practices, it is essential to explain why certain datasets are unavailable. According to W3C, data consumers need to know when "data that is referred to from the current dataset is unavailable or only available under different conditions" (Lóscio et al., 2017, Best Practice 22). An example from the W3C best practices is that in the case of datasets containing bus stops information, the dataset "can contain sensitive data, for instance, information about the bus driver. In this case, the publisher provides an explanation informing potential users that the personal data about the bus driver is not available." This particular best practice is linked with the "reuse" and "trust" aspects in the W3C metrics of benefits.

All of these suggest that the gaps in data policy statements could lead to serious problems in citizens' use of open government data. These serious problems could turn into major obstacles, especially when it comes to emergency public health situations. During the lockdown periods of the COVID-19 pandemic, citizens turn to their government for accurate and timely information. As cited earlier, Curioso and Carrasco-Escobar (2020) assert that providing open data is critical to "face this pandemic," and the government needs to ensure transparency and data-driven decision-making. The gaps that we identified through our investigation in dashboard data policy, when aggregated together, could lead to going against the very core of open data and making citizens not use or not trust the open data. Our recommendations in the previous section might be a first step towards developing a data policy statement that matches the fundamental principles of open data and thereby benefits citizens or public health professionals who use dashboards to assess the situation and get informed to navigate their everyday life. Furthermore, the best practice examples provided by W3C in their "Data on the Web Best Practices" can give concrete actionable items to address the notable gaps listed here. For instance, several W3C's suggestions such as "Provide descriptive and structural metadata," "Provide data license information," "Provide a version history," "Use persistent URIs as identifiers of datasets" and "Provide an explanation for data that is not available" are all related to improving the "reusable" principle and may address the observed problems in "Data source," "Historical/Archived datasets," "Data subject to change" and "Data suppression disclaimer."

Table 2
Mapping of inconsistencies/gaps to open data principles and FAIR guideline.

Data Policy Elements	Inconsistencies/Gaps	Impact as per Open Data Principles, *Problem with:*	Impact as per FAIR, *Problem with*
Data source	● Missing original data sources where the state collected data from	Described Reusable Complete	Findable Reusable
Data update frequency	● Inconsistent placement ● Missing update frequency for specific sections	Timely Complete	Findable Interoperable
Historical/ Archived datasets	● Absence of description of or link to archived datasets; ● Inconsistency in the format of archived datasets	Public Accessible Described Managed post-release	Findable Accessible Interoperable Reusable
Definitions of terms	● Inconsistent placement ● Definitions vary by depth and breadth ● Original source of definitions not provided	Described Complete	Findable Interoperable
Data scope	● Data coverage not specified	Described Reusable	Reusable
Data subject to change disclaimer	● Not all dashboards have the "subject to change" disclaimer	Described Reusable	Reusable
Community data level	● Community risk level not explicitly stated whether it follows or refers to CDC's	Described	Interoperable Reusable
Data Suppression Disclaimer	● Suppression threshold unclear ● Suppression outcome to data counts unclear	Described Reusable Complete	Interoperable Reusable

To improve the open data provision, establishing a consistent and standardized data policy is crucial. Although we identified a number of gaps, there are actionable areas where government agencies can work on to provide stewardship by harmonizing requirements for data dashboards or other open data portals to publish their datasets. As an example, the European Union's "data publication guide" presents an effort to "structure," "write," "visualize," "prepare visuals," "prepare data," and "publish data." (Publications Office of the European Union, n.d.). As a result of this effort, Eurostat implemented a data quality control procedure harmonizing metadata in the field of official statistics across EU's members. Another best practice example is the Cooperation Open Government Data Austria's "Framework for Open Government Data platforms", which defines the "legal, technical and organizational requirements for Open Government Data platforms" (Eibl et al., 2019, p. 1). Austria's "Data Publication Guide" is also an example of the government's leadership in establishing and activating data policies to ensure consistency and standardization in the open data quality.

(Cooperation OGD Austria, n.d.). Within the US, nevertheless, there does not seem to be legal principles in place for open data policy beyond the high level principles or W3C best practices. If such requirements are in place for federal, state and other local government open data, the gaps that we observed in our study might be reduced to a significant extent.

In conclusion, our study focuses only on the U.S. state dashboards; as such, our findings might not be generalizable to COVID-19 dashboard data policies around the world. Nevertheless, both our analysis model and our approach of linking ODP and the FAIR principles with our investigation of gaps and inconsistencies in data policy statements are a fruitful direction, so is our discussion on the potential government mandated open data publishing requirements that could lead to the reduction of those gaps. The analysis model may be applicable if we expand the scope of our study to the global level. Such a model might also be applicable to assess data use policies published on OGD sites. Further research may be carried out to examine the data notes published on all government sites containing open datasets. It is our belief that merely providing open datasets is not sufficient, and when developing data use policy statements, it is essential to follow the ODP and the FAIR principles to make sure all elements are included without major gaps or errors. Furthermore, as shown through the example of EU and Austria, the leadership and effort from a government in establishing and implementing open government data publishing standards, rooted on the ODP, FAIR Principles, and W3C best practice examples, may help to make significant progress in this arena by reducing the observed gaps and errors in open data use policy statements, and thereby improving open data provision and beyond.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Cay, D., Nagel, T., & Yantaç, A. E. (2020). Understanding user experience of COVID-19 maps through remote elicitation interviews. In *2020 IEEE workshop on evaluation and beyond-methodological approaches to visualization (BELIV)* (pp. 65–73). IEEE. https://doi.org/10.1109/BELIV51497.2020.00015.

Cooperation OGD Austria. Data publication guide (n.d.) https://www.data.gv.at/en/data/data-publication-guide/.

Curioso, W. H., & Carrasco-Escobar, G. (2020). Collaboration in times of COVID-19: The urgent need for open-data sharing in Latin America. *BMJ Health & Care Informatics, 27*(1), Article e100159. https://doi.org/10.1136/bmjhci-2020-100159

Davies, T. (2010). *Open data, democracy and public sector reform*. Retrieved from http://www.practicalparticipation.co.uk/odi/report/.

Dombrowski, Q. (2022). Review: jsLDA. *Reviews in Digital Humanities, III*(8). https://reviewsindh.pubpub.org/pub/jslda.

Eibl, G., Lutz, B., Parycek, P., & Pawel, S. (2019). *Framework for open government data platforms*. Cooperation Open Government Data Austria. https://go.gv.at/ogdframeen.

Goodspeed, R. (2011). From public records to open government: Access to Massachusetts municipal geographic data. *URISA Journal, 23*(2), 21–32.

Horrigan, J., & Rainie, L. (2015). *Americans' views on open government data*. Pew Research Center. Retrieved from https://www.pewresearch.org/internet/2015/04/21/open-government-data/.

jsLDA: In-browser topic modeling (n.d.). Retrieved from https://mimno.infosci.cornell.edu/jsLDA/jslda.html.

Klein, F. L., Eisenstein, J., & Sun, I. (2015). Exploratory thematic analysis for digitized archival collections. *Digital Scholarship in the Humanities, 30*(Supplement 1), i130–i141. https://doi.org/10.1093/llc/fqv052

Lóscio, B. F., Burle, C., & Calegari, N. (2017). *Data on the web best practices*. W3C Recommendation 31 January 2017 https://www.w3.org/TR/dwbp/.

Matheus, R., Janssen, M., & Maheshwari, D. (2020). Data science empowering the public: Data-driven dashboards for transparent and accountable decision-making in smart cities. *Government Information Quarterly, 37*(3). https://doi.org/10.1016/j.giq.2018.01.006

Mimno, D. (2013). *jsLDA License*. https://github.com/mimno/jsLDA/blob/master/LICENSE.

Nayek, J. K. (2018). Evaluation of open data government sites: A comparative study. *Library Philosophy and Practice, 1781*. Retrieved from https://digitalcommons.unl.edu/libphilprac/1781.

Project Open Data. (2010). *Open data principles*. Retrieved from https://resources.data.gov/PoD/principles/.

Publications Office of European Union. (n.d.) A practical guide to user-friendly data in publications. https://data.europa.eu/apps/data-in-publications-guide/ .

Röddiger, T., Beigl, M., Dörner, D., & Budde, M. (2020). Responsible, automated data gathering for timely citizen dashboard provision during a global pandemic (covid-19). *Digital Government: Research and Practice, 2*(1), 1–9. https://doi.org/10.1145/3428471

Shah, N. (2020). *Data visualization and dashboards in the era of Covid-19*. Retrieved from https://opendatascience.com/data-visualization-and-dashboards-in-the-era-of-covid-19/.

Sinclair, S., & Rockwell, G. (2016). *Voyant tools. Web*. http://voyant-tools.org/.

Smith, A. (2010). Government Online: The internet gives citizens new paths to government services and information. *Pew Research Center*. Retrieved from https://www.pewresearch.org/internet/2010/04/27/government-online/.

Tang, R., & Jiang, J. (2021). Characteristics of open government data (OGD) around the world: A country-based comparative meta-analysis. *Data and Information Management, 5*(1), 11–26. https://doi.org/10.2478/dim-2020-0026

Tauberer, J.. The 8 principles of open government data (n.d.) https://opengovdata.org/.

Voyant Tools. (n.d.). *About*. https://voyant-tools.org/docs/#!/guide/about.

Wang, V., Shepherd, D., & Button, M. (2019). The barriers to the opening of government data in the UK: A view from the bottom. *Information Polity, 24*(1), 59–74. https://doi.org/10.3233/IP-180107

White House. (2009). *Memorandum on transparency and open government*. Retrieved from https://www.archives.gov/files/cui/documents/2009-WH-memo-on-transparency-and-open-government.pdf.

Wilkinson, M., Dumontier, M., Aalbersberg, I., et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. *Scientific Data, 3*, Article 160018. https://doi.org/10.1038/sdata.2016.18

Zuiderwijk, A., & Janssen, M. (2014). Open data policies, their implementation and impact: A framework for comparison. *Government Information Quarterly, 31*(1), 17–29.

Zuiderwijk, A., & Reuver, M. de (2021). Why open government data initiatives fail to achieve their objectives: Categorizing and prioritizing barriers through a global survey. *Transforming Government: People, Process and Policy, 15*(4), 377–395. https://doi.org/10.1108/TG-09-2020-0271